JIN-BIN, IM – VISITING RESEARCHER

Chalmers University of Technology, Department of Architecture and Civil Engineering
Division of Construction Management and Engineering
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PROFILE

Architectural researcher and designer who turns how people describe the spaces they need into machine-readable building models and tests the designs with their future occupants. The work combines NLP and large language models, BIM automation, and user studies in mixed and virtual reality.

RESEARCH INTERESTS

AI-Driven Architectural Design & Computing: **A**ffective computing for spatial design generation; Client requirements-driven automated design optimization; Evidence-based decision support systems for personalized architectural design; Enhancing communication quality between architect and laypeople

Immersive Virtual Environment for Occupant Assessment: XR (VR/MR) platforms for immersive co-design and real-time spatial experience with clients; Human–space interaction and occupant behavior analysis in dynamic architectural environments; Sensor-driven evidence-based design for occupant response assessment

Human-Centered Digital Design Workflow: End-to-end pipelines from multimodal data acquisition to BIM-linked structured output; Natural language processing for translating user requirements into design parameters; Interoperable digital design frameworks supporting downstream lifecycle analysis

EDUCATION

Aug 2026 — **Hanyang University, Seoul, South Korea**
M.S./Ph.D. in Architectural Engineering *Advisor: Ju-Hyung Kim, JoonOh Seo*

Feb 2020 — **Hanyang University, Seoul, South Korea**
B.S. *in Interior Architecture Design*

EXPERIENCE

09/2026 – present — **Visiting Researcher**, Chalmers University of Technology
Host: Prof. Mattias Roupé

2020 – 2026 — **Researcher,** Hanyang University
Advisor: Ju-Hyung Kim

2019 – 2020 — **Intern, property management**, SIFC Property Korea YH
Task: operations of the IFC Seoul complex and IFC Mall (85,444 m² GFA)

2019 — **Research intern**, Hanyang University
Mentor: Seung Hyun Cha

TEACHING

2026 — **Instructor** (online course), Korea University of Technology and Education
Course: Generative AI for Architectural Design Optimization

Fall 2025 — **Teaching Fellow**, Department of Architectural Engineering, Hanyang University
Course: Architectural Engineering Capstone PBL

Spring 2024, 2025 — **Teaching Fellow**, Department of Architectural Engineering, Hanyang University
Course: Computer Engineering

RESEARCH PROJECTS

2026 **Development of multi-functional construction robots to enhance productivity in apartment housing construction**
Role: *Research Fellow at the CB&PM Lab. (Hanyang University)*
Funding: *Ministry of Land, Infrastructure and Transport*

2024 – 2026 **Development of an AI-based real-time construction quality management system**
Role: *Research Fellow at the CB&PM Lab. (Hanyang University)*
Funding: *National Research Foundation of Korea*

2023 – 2024 **Quality control of educational facilities using virtual environments**
Role: *Lead Researcher at the CB&PM Lab. (Hanyang University)*
Funding: *Poongjak Co., Ltd*

2021 – 2024 **Decision-making framework for personalised spatial composition based on temperament, perception and bio-markers, applying digital twins.**
Role: *Lead Researcher at the CB&PM Lab. (Hanyang University)*
Funding: *National Research Foundation of Korea*

2021 – 2022 **Construction quality control using digital twin technology**
Role: *Research Assistant at the CB&PM Lab. (Hanyang University)*
Funding: *Hyundai Engineering Co., Ltd.*

2021 **Standard guidelines for daycare centres in police stations**
Role: *Research Assistant at the CB&PM Lab. (Hanyang University)*
Funding: *Korean National Police Agency*

2019 **Spatial preference modeling based on immersive virtual environment and choice modeling theory**
Role: *Research Intern at Dept. of Interior Architecture Design (Hanyang University)*
Funding: *National Research Foundation of Korea*

ACADEMIC SERVICE

2026 – present **Reviewer, Virtual Reality**, *Springer Nature*.

2026 – present **Reviewer, Acta Psychologica**, *Elsevier*.

2026 – present **Reviewer, Results in Engineering**, *Elsevier*.

2025 – present **Reviewer, Journal of Management in Engineering**, *ASCE*.

2025 – present **Reviewer, Architectural Engineering and Design Management**, *Taylor & Francis*.

2025 – present **Reviewer, Building and Environment**, *Elsevier*.

PATENT

2022 Republic of Korea Patent No. 10-2421360, *System for Generating 3D LiDAR Scanning Position*

PUBLICATIONS

Bold = applicant; * = corresponding author.

- ### REFEREED INTERNATIONAL JOURNALS

[J15] Zhang, E., **Im, J. B.,** Lee, K. T., & Kim, J. H. * (2026). Visual quality features in immersive virtual environments: A systematic review of their impact on user experience. *Virtual Reality*, Springer Nature. (accepted) https://doi.org/10.1007/s10055-026-01433-z

[J14] Hong, R. L., Yoon, J. H., **Im, J. B.,** & Kim, J. H. * (2026). Prediction of actual safety and health management expenses for water supply system construction projects in Korea through machine learning. *Journal of Civil Engineering and Management*. (accepted)

[J13] Hong, R. L., **Im, J. B.,** Kang, H. G., & Kim, J. H. * (2026). Prediction of structural frame works safety and health management expenses in apartment construction projects by introducing interpretable machine learning. *Engineering, Construction and Architectural Management*. https://doi.org/10.1108/ECAM-07-2025-1073

[J12] Joo, M., **Im, J. B.,** Lee, K. T., & Kim, J. H. * (2026). Comparative analysis of work performance and physiological response based on office types and indoor space design. *Architectural Engineering and Design Management*, 22(2), 365–381. https://doi.org/10.1080/17452007.2025.2490958

[J11] Lee, K. T., **Im, J. B.,** & Kim, J. H. * (2026). Effects of human perception dimensions constituting individual temperament and experience on indoor space composition decision-making presented in a mixed-reality environment. *Architectural Science Review*, 69(3), 314–329. https://doi.org/10.1080/00038628.2025.2510300

[J10] Lee, K. T., Shin, S. H., **Im, J. B.,** Choi, H., & Kim, J. H. * (2026). Integrating AI-enhanced and conventional market selection approaches for the construction industry: Insights from Indian firms. *Journal of Civil Engineering and Management*, 32(3), 456–474. https://doi.org/10.3846/jcem.2026.26811

[J9] Lee, K. T., **Im, J. B.,** Hong, R. L., & Kim, J. H.* (2026). Enhancing retrieval-augmented generation performance through network analysis of question types in public procurement law. *Journal of Management in Engineering*, ASCE, 42(4), 04026014. https://doi.org/10.1061/JMENEA.MEENG-7429

[J8] **Im, J. B.,** Hong, R. L., Joo, M., Zhang, E., & Kim, J. H. * (2025). Visual fidelity effects on occupants' performance, mental states, and emotions in mixed reality environments. *Architectural Science Review*, 1–21. https://doi.org/10.1080/00038628.2025.2584180

[J7] **Im, J. B.,** Joo, M., Lee, K. T., & Kim, J. H. * (2025). A mixed reality approach to identify cognitive performance and mental states in preferred vs. non-preferred individual and collaborative work environments. *Building and Environment*, 274, 112806. https://doi.org/10.1016/j.buildenv.2025.112806

[J6] Park, S. J., **Im, J. B.,** Yoon, H. S., & Kim, J. H. * (2025). Data augmentation approaches for estimating curtain wall construction duration in high-rise buildings. *Buildings*, 15(4), 583. https://doi.org/10.3390/buildings15040583

[J5] Kim, H. J., Kim, J. H.*, & **Im, J. B.,*** (2023). Forecasting offline retail sales in the COVID-19 pandemic period: A case study of a complex shopping mall in South Korea. *Buildings*, 13(3), 627. https://doi.org/10.3390/buildings13030627

[J4] Lee, K. T., **Im, J. B.,** Park, S. J., & Kim, J. H. * (2023). Identifying improvement priorities in daycare centre space performance from teachers' perspectives: Application of importance–performance analysis and structural equation modelling. *Engineering, Construction and Architectural Management*, 32(1), 639–672. https://doi.org/10.1108/ECAM-08-2022-0777

[J3] Lee, K. T., **Im, J. B.,** Park, S. J., & Kim, J. H. * (2022). Conceptual framework to support personalized indoor space design decision-making: A systematic literature review. *Buildings*, 12(6), 716. https://doi.org/10.3390/buildings12060716

[J2] Park, S. J., Lee, K. T., **Im, J. B.,** & Kim, J. H.* (2022). The need for smart architecture caused by the impact of COVID-19 upon architecture and city: A systematic literature review. *Sustainability*, 14(13), 7900. https://doi.org/10.3390/su14137900

[J1] Nindartin, A., Moon, H. W., Park, S. J., Lee, K. T., **Im, J. B.,*** & Kim, J. H.* (2022). Influencing of the building energy policies upon the efficiency of energy consumption: The case of courthouse buildings in South Korea. *Energies*, 15(18), 6679. https://doi.org/10.3390/en15186679

- ## REFEREED INTERNATIONAL CONFERENCES

[C11] Hong, R. L., **Im, J. B.,** Shim, W. S., Wang, S. H., Ha, W. C., & Kim, J. H. * (2026). Automatic semantic parsing of structural drawings and rebar schedules based on a hybrid deep learning framework. Proc. 43rd ISARC, Singapore. https://doi.org/10.22260/ISARC2026/0197

[C10] **Im, J. B.,** Hong, R. L., Choi, C. H., Ha, J. E., & Kim, J. H. * (2026). Optimal transport-based affective user requirements realignment for automated minimal disruptive spatial design revision in BIM environment. Proc. 43rd ISARC, Singapore. https://doi.org/10.22260/ISARC2026/0251 (presenter)

[C9] **Im, J. B.,** Ye, S. J., Zhang, E., Lee, K. M., Lee, K. T., & Kim, J. H. * (2026). Assessing material and labor cost indices impacts on construction firms' financial performance: A machine learning comparison with feature engineering. Proc. 43rd ISARC, Singapore. https://doi.org/10.22260/ISARC2026/0107

[C8] Zhang, E., **Im, J. B.,** Xu, L., Park, S. Y., & Kim, J. H. * (2026). The influence of window characteristics on occupant well-being and perception in indoor environments: A review. Proc. 12th IAQVEC, Los Angeles. https://doi.org/10.1051/e3sconf/202671605006

[C7] Zhang, E., Choi, S. W., **Im, J. B.,** Baek, Y. G., Kim, J. H., & Lee, K. T. * (2025). Construction conflict resolution strategies focused on two-level game theory: A case study analysis. Proc. 11th ICCEPM, Jeju. https://doi.org/10.6106/ICCEPM.2025.0152

[C6] Ochirsuren, N., Kim, G. T., **Im, J. B.,** Hong, R. L., Xu, L., & Kim, J. H. * (2025). A BPMN-based framework for real-time rebar arrangement inspection on construction sites. Proc. 11th ICCEPM, Jeju, 136–143. https://doi.org/10.6106/ICCEPM.2025.0136

[C5] Hong, R. L., **Im, J. B.,** Zhang, E., et al., & Kim, J. H. * (2025). Development of a petrochemical plant pipe counting method based on improved Mask R-CNN and Transformer. Proc. 42nd ISARC, Montreal, 972–979. https://doi.org/10.22260/ISARC2025/0126

[C4] **Im, J. B.,** Hong, R. L., Jang, J. H., Zhang, E., et al., & Kim, J. H. * (2025). Effects of visual fidelity on task performance and mental workload in IVEs. Proc. 42nd ISARC, Montreal, 909–916. https://doi.org/10.22260/ISARC2025/0118 (presenter)

[C3] Jang, J. H., **Im, J. B.,*** Zhang, E., Joo, M., Kang, S. H., & Kim, J. H. * (2024). A framework for assessing the learning performance and creativity in spatial features by immersive virtual environments. Proc. 10th ICCEPM, Sapporo, 621–628. https://doi.org/10.6106/ICCEPM.2024.0621 (presenter)

[C2] Park, S. J., Lee, K. T., **Im, J. B.,** & Kim, J. H. * (2022). Smart architecture in response to pandemic: A systematic literature review. Proc. 13th ISAIA, Wuhan.

[C1] Lee, K. T., Park, S. J., **Im, J. B.,** & Kim, J. H. * (2022). Hierarchical organization structure framework for negotiation consensus in multinational joint-client projects. Proc. 13th ISAIA, Wuhan, 133–140.

- ## REFEREED DOMESTIC JOURNALS

[D6] Shin, S. H., Lee, K. M., **Im, J. B.,** & Kim, J. H. * (2026). Factor analysis on cost overrun by introducing FMEA and proposal for improving approximate estimation reliability in apartment projects. Journal of the Architectural Institute of Korea, 42(3), 339–347. https://doi.org/10.5659/JAIK.2026.42.3.339

[D5] Zhang, E., Jo, S. H., **Im, J. B.,** & Kim, J. H. * (2025). Importance and performance analysis on stakeholders' perceptions against public rental housing status and directions to improve occupants' satisfaction. Journal of the Architectural Institute of Korea, 41(1), 301–312. https://doi.org/10.5659/JAIK.2025.41.1.301

[D4] Lee, K. Y., Ding, Z. S., **Im, J. B.,** Park, S. J., & Kim, J. H. * (2024). Investigation of construction duration influence factors and comparison of artificial intelligence methods for estimating appropriate construction duration in public sewerage construction. Korean Journal of Construction Engineering and Management, 25(6), 66–74. https://dx.doi.org/10.6106/KJCEM.2024.25.6.066

[D3] Kim, D. H., Ye, S. J., Lee, K. T., **Im, J. B.,** & Kim, J. H. * (2023). Developing a comprehensive framework for detailed facility planning in office building grade assessment. Journal of the Architectural Institute of Korea, 39(9), 69–80. https://doi.org/10.5659/JAIK.2023.39.9.69

[D2] **Im, J. B.,** Hwang, S. J., Jang, J. H., Kim, J. H., & Lee, K. T. * (2022). Introduction of mixed reality on special tree landscape design review phase. Journal of the Architectural Institute of Korea, 38(8), 71–81. https://doi.org/10.5659/JAIK.2022.38.8.71

[D1] Kim, H. S., **Im, J. B.,** Lee, K. T., Kim, H. J., & Kim, J. H.* (2021). Proposals for feasibility study and optimal combination of renewable energy generation systems in Baengnyeongdo and Oedo using HOMER. Journal of the Architectural Institute of Korea, 37(12), 347–356. https://doi.org/10.5659/JAIK.2021.37.12.347

- ## ONGOING RESEARCH

[M1] Strategic pathways for e-methanol deployment in the low-carbon transition: Policy modeling of cost competitiveness and flexibility requirements in Korea. *Under revision, Energy Strategy Reviews.*

[M2] Beyond viewing: How installation artwork design modality shapes visitor satisfaction through affective and behavioral pathways. *Under revision, the Design Journal.*

[M3] Weather-adaptive automated pipe counting for open-air construction storage: Leveraging synthetic data augmentation and YOLO-network family. *Under revision, Scientific Reports.*

[M4] How much realism is enough? Mixed and virtual reality interiors versus the real room. *Under review, Journal of Interior Design.*

[M5] Quantifying affect as a design requirement: An affective engineering approach to modelling multi-dimensional emotional response of residential interiors. *Manuscript completed.*

[M6] From 2D drawings to structured inspection data: A scale-aware framework for rebar information parsing via SAHI-based YOLO, OCR, and LLM. *Manuscript completed.*

[M7] External bargaining and internal ratification in Seoul urban redevelopment through two-level game theory.
Manuscript completed.

[M8] Automated BIM design revision from residents' natural-language emotional feedback. *Manuscript completed.*

[M9] Management backing and the perceived benefits of a mandatory supplier-management system in EPC procurement. *Manuscript completed.*

[M10] Design intent in generative AI for architectural design: Actors, requirements, input representation, and intent fulfilment. *In preparation.*

[M11] Enacted activity shapes the affective evaluation of the same interior in virtual reality. *In preparation.*

REFERENCES

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